Course architecture

Advanced Statistics 2027-1 / Probabilistic Machine Learning · Course-level intellectual and curricular authority

This document defines the **intellectual purpose and planned progression** of the course. It is the primary reference when selecting lecture scopes, weighing changes across lectures, and interpreting the relationship between mathematical results and applications. The student syllabus is a concise description of this architecture; approved lecture scopes determine their own detailed teaching commitments. Actual delivery and source coverage are recorded in the material ledger. These authorities should be reconciled when they disagree, not silently overwritten.

What the course is about

Understanding probabilistic machine learning is understanding **the probabalistic formulation of machine language problems**, **how different machine language problems are expressed in this machinery**, and **why specific classes of learning problems apply different computational approaches to the same machinery**. The goal is to achieve a coherent and integrated view of how the structured data, the questions being asked, and the computational tools are all related through the probabalistic formulation. This will not be a catalogue of models and it is not a progression from one approximation algorithm to the next. Each mathematical advance is motivated by a kind of task and available experience for which our existing models, performance criteria, or computations are insufficient.

From learning problems to mathematical formulations

A **learning task** specifies what a system should accomplish; its **experience** specifies the data, feedback, or interaction available. Its success must be defined by a **performance criterion**. The same dataset may support prediction, generation, discovery, or action; data alone do not specify the learning problem.

The mathematical formulation has **two distinct, related objects**. A probabilistic model describes the relevant observed and unobserved variables, their dependencies, and our uncertainty. A performance functional defines what counts as a good prediction, representation, decision, or learned distribution—often by taking an expectation with respect to a relevant distribution. For example, a predictive action a may be evaluated by

$$\mathcal{R}(a) = \int L(a, y)p(y)\,\mathrm{d}y.$$

The distribution under which we evaluate actual performance need not be the same as our fitted model. Distinguishing a mismatch of models from an error of computation is itself part of the discipline.

The formulation then gives us probabilistic **queries**. These are generally addressed with specific integrals on the probabilistic models: conditioning, marginalization, and summarizing are three operations developed explicitly that may deliver the quantities required to evaluate or optimize performance. They are tractable in some models, but the course deals primarily with how queries can be approached when computation is a challenge.

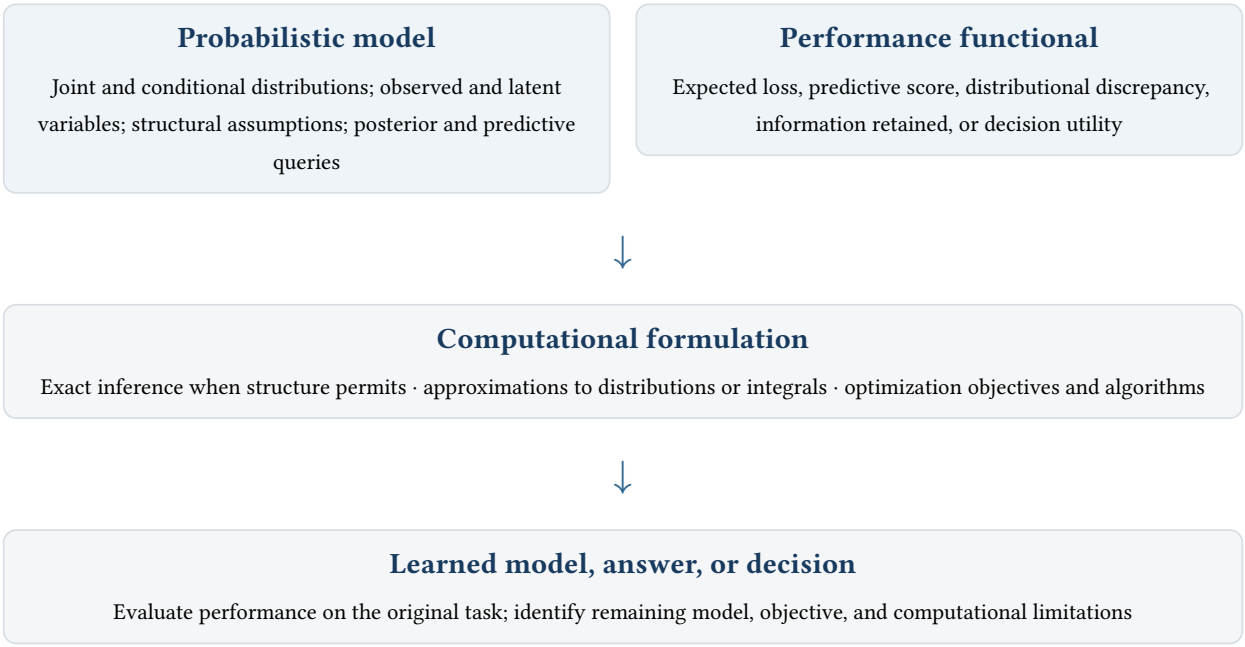
The task–model–performance–computation relationship

Learning problem

Task to be accomplished · available experience and data structure

↓

Mathematical formulation



The arrows describe the logical dependence, not a one-way workflow. Results and computational limits can motivate revising the formulation.

The three major **moves** in the course are consequently extensions of **(i) what models can represent, (ii) what performance functionals can express, and (iii) what computation can achieve**. A task family motivates one move or a combination; no move is an end in itself.

The ideal quantity and the practical calculation

One key move in the course is to substitue a tractable proxy for an ideal mathematical target. A desired performance functional may be replaced during training by an empirical score or surrogate objective. Likewise, an ideal posterior or expectation may be represented by a point estimate, a local Gaussian approximation, a variational distribution, or samples. Some structured problems admit **exact** solutions, so computation is not synonymous with approximation.

The central goal of the course is to develop an understanding of the possible substitutions and the situations in which they are appropriate. Each substitution has different meanings and limitations. The appropriate criterion for assessing the approximation depends on the **original question**, **the probabilistic formulation**, and **the computational power available** not merely whether an internal algorithm converged.

The role of Murphy and mathematical depth

Murphy's books provide the central spine of the course's progression and guide the choice of material, the explanatory logic, the depth of the coverage, and the mathematical language. The course as a whole and each lecture provides only partial coverage, relative to Murphy's second book. Material is selected and occasionally re-ordered.

A detailed understanding of the mathematical development of the material is an essential part of the course. As such, many of the mathematical developments in Murphy are presented in lecture or in the self-work notebooks. The first Murphy book often provides more detailed and scaffolded mathematical derivations, and is used as a reference for the course. The second Murphy book is the primary source fo the underlying logic of the course (as explained above) and for its structure.

How the semester develops

Each lecture develops the models, mathematical formulations, and computational tools to address a class of problems. Thus, each lecture can be seen as a discrete move extending the classes of problems that can be

addressed and the interaction between problems, formulation, and computation. Each lecture should reach a substantive conclusion and make the move it made and the achievment that enables explicit. Each lecture is followed by a notebook that extends the lecture with a worked example, usualy illustrating the type of problem that can now be solved and exercising the computational approach.

Foundations: questions, distributions, and structure

L01 defines machine learning andestablishes the framework outlined in this document. The framework is made concrete using Linear-Gaussian systems and specific computational tools are applied to demonstrate how conditioning, marginalization, and summarizing work together to solve an estimation problem.
L02 extends the class of probabilistic models we consider by developing the mathematics of the exponential families. It also introduces derives fitting by moment matching; learning categorical transitions leads to a simple sequence generator.
L03 examines factorizations and conditional independence to express larger structured joint models without writing a full joint table.

Richer performance criteria and tractable learning

L04 addresses tasks for which single-answer classification accuracy or mean squared error is inadequate. Distributional similarity, relative information, and preservation of relevant statistical relationships motivate entropy, cross-entropy, KL divergence, mutual information, and their relationship to computable scores or bounds. The selected small noisy-channel example is a **worked illustration**, not the intellectual definition of the chapter or a representative machine-learning application: it distinguishes information, decision reliability, and resource cost. The three-bit detector is live; the five-bit correction problem extends it in the notebook.
L05 connects familiar Bayesian regression to likelihood families for categorical and count observations. Exponential-family GLMs extend the **model**. Differentiation and stochastic optimization address the practical problem of **learning** from finite observations. Using MLE or MAP as a point fit is a computational choice that retains the probabilistic model but not a full parameter posterior. Prediction is ultimately judged by an appropriate performance criterion; success on an internal training objective is not equivalent to task performance.
L06 retains a familiar latent-variable hierarchy and asks how parameters can be learned when the observed-data likelihood involves marginalization over hidden values. EM uses conditional inference to construct tractable expected-statistic fitting steps, under explicit assumptions. The same probabilistic framework motivates the method; it is not a competing definition of hierarchical modeling.

Inference and computation under different structural conditions

L07 returns to the difference between a point fit and posterior queries that require uncertainty or predictive integration. Laplace uses local curvature for a Gaussian approximation; the approximation must be checked for the query that matters.
L08 addresses sequential tasks where the naive latent sum appears exponential but the joint factorization enables efficient **exact** filtering and smoothing by message passing. **L09** instead optimizes a restricted distribution to approximate an otherwise difficult posterior; **L10** uses samples to approximate integrals and studies the validity and finite-run error of Monte Carlo estimates. These methods solve different computational bottlenecks, not interchangeable names for “approximate inference.”

Expressive models and learned inference

L11 develops compositional differentiable functions as components of conditional probability models. Greater expressive capacity can improve nonlinear prediction but does not itself solve parameter uncertainty. Neural

networks can also parameterize **inference** maps, a role developed in **L12**. The VAE combines a generative latent-variable model, amortized posterior approximation, and an objective that trades reconstruction quality against an information-related rate. Generation and discovery are different tasks even when they share the same model.

Lecture allocation and notebook commitments

The table below retains the selected 13-meeting allocation and source readings. It is a **planning contract**: L01 is delivered, L02 is agreed, L03 preserves its instructor-approved scope, and L04′s live/notebook split is selected. Other lecture scopes remain proposed pending their own reviews. Each notebook should extend rather than reproduce the live mathematics, normally within a separate hour. Full mathematical source-to-lecture ownership and unresolved feasibility questions remain in the architecture audit and the scope register.

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook
1	**Probabilistic ML and Gaussian systems** Delivered	Machine-learning task framing; probabilistic models; Bayes review; covariance and correlation; multivariate Gaussians; conditioning; missing-data inference; linear Gaussian systems.	• 2.1.5–2.1.6 Conditional probability; Bayes • 2.2.1.1 Bernoulli and binomial • 2.2.4.1 Beta distribution • 2.3.1.1 Gaussian joint distribution • 2.3.1.3–2.3.1.5 Gaussian conditionals; information form • 2.3.2.1–2.3.2.2 Linear-Gaussian systems • 3.2.1 Coin-flipping Bayes • 3.4–3.4.1 Conjugate priors; binomial	• 1.1–1.5 ML tasks and datasets • 2.3 Bayes' rule • 2.4.1 Bernoulli; binomial • 2.7.4 Beta distribution • 3.1.1–3.1.3 Covariance; correlation • 3.2.1–3.3.4 Multivariate and linear Gaussians • 4.3–4.3.1 Empirical risk • 4.6.1–4.6.2 Beta-binomial Bayes	**Gaussian** **delivered** observed uncertaint
2	**Exponential families and learning from sufficient statistics** Agreed scope	Exponential-family representation; Gaussian canonical form; derivatives of the log-partition function; natural and mean parameters; MLE as moment matching; maximum entropy; Gamma fit; categorical counts; Markov-chain learning.	• 2.2.1.2 Categorical distribution • 2.4.1–2.4.2 Exponential-family form; examples • 2.4.3–2.4.4 Log partition; moments • 2.4.5 MLE; moment matching • 2.4.7 Maximum entropy • 2.6.1–2.6.3 Markov chains; language models		**Baby-nam** transitions matrix, co Reference
3	**Graphical models: factorization and conditional independence** Agreed scope restored	Derive DAG factorization from the probability chain rule and independence assumptions; work the student network and parameter economy. Develop d-separation and directed Markov blankets. Define undirected potentials and global normalization, compare separation and independence expressivity, and interpret the binary Ising model. Image posterior and algorithm derivations belong to the notebook.	• 4.2.1–4.2.2 DAG factorization; student network • 4.2.4 / selected 4.2.8 Independence; plates • 4.3.1–4.3.2.1 / 4.3.6.1 Undirected models; Ising; separation • 4.5.1 Directed/undirected independence comparison • 4.3.8 Image restoration: notebook	• Optional 3.6.1–3.6.2 Graphical-model orientation	Construct likelihood conditiona maximizat pixelwise optimum restoration update, an optimum source. **Li** method. It promise th All course
4	**Information theory: mutual information, processing and noisy coding** Proposed teaching scope	Finite entropy, cross-entropy and KL divergence; derive mutual information and sketch the data-processing inequality with its conditional-independence premise. State source-length bounds. Work three-bit parity detection and posterior decoding, distinguishing information, Bayes message risk and a coding guarantee.	• 5.1.3 / 5.1.6; 5.2.1 / 5.2.4 Information and uncertainty • 5.3.1–5.3.4 Shared information and processing • 5.4.1 Code-length bounds: statement only • 5.5 Noisy-channel example: three-bit parity model	• Selected 6.1 / 6.2.1–6.2.2 / 6.3 Uncertainty and shared information	Extend the correcting minimum bit flip; us compute and compa benchmar informatio distributio and guara The lectur notebook Guarantee notebook untested.
5	**Generalized linear models, automatic differentiation and SGD** Proposed teaching scope	Start from the familiar Bayesian regression likelihood and parameter prior; explain why Gaussian observations are unsuitable for binary responses and counts, without implying that the hierarchical model fails. Generalize the observation family using Gaussian/Bernoulli/Poisson GLMs and links. Contrast full posterior inference with MLE/MAP point fitting; derive the canonical NLL and vector softmax gradients, then reverse-mode autodiff and minibatch SGD.	• 15.1 / 15.1.1 / 15.1.3 Common model structure and derivative • 15.1.2 Brief alternative-link comparison • 15.3.2 Three-class probabilities • Selected 6.2; 6.3.1–6.3.2 Automatic differentiation and batch updates	• Selected 7.8.2 / 7.8.5 Matrix derivatives • Selected 8.2 / 8.4.1 / 8.4.3 Gradient and batch-update support	Fit the thr the manua parameter log loss/pr scaffolding derivative a checked Inspected meeting ti the lecture course ada
6	**Latent-variable learning and expectation-maximization** Proposed teaching scope	Recall Bayesian latent-variable hierarchies and joint posterior inference, then distinguish point estimation by marginal likelihood from inference over all unknowns. Show why unknown latent values complicate direct fitting, and derive the ELBO/KL identity, exact E-step tightness and EM monotonicity. Apply Gaussian conditional first/second moments to unknown mean/covariance with missing data; briefly interpret mixture responsibilities and assumptions.	• Selected 6.5.1 / 6.5.3 Probability bound and EM • 6.5.4 Gaussian missing-data updates • 3.11 When the missing-data mechanism can be ignored • 28.2.1 Group-assignment probabilities: brief example	• Selected 3.5 / 8.7 Grouped observations and EM explanation	Assemble conditiona a supplied missing-e **Skill:** Use estimation guarantee updates. I candidate probabiliti fit is not r
7	**Bayesian computation: grid reference and Laplace**	Posterior queries and parameterization-dependent MAP; derive scalar Laplace curvature and normalizer under positive-curvature conditions. Apply to proper-prior	• 7.1–7.3 What posterior questions require: short introduction		Build a sta observatio separately prediction

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook
	approximation Proposed teaching scope	logistic regression, check quadrature domain and resolution, and compare posterior moments and predictive integrals.	• Selected 7.4.1–7.4.3 Most likely parameter, grid and Gaussian shortcut • Selected 7.5 Checking approximation quality • Selected 15.3.5–15.3.6 One-parameter logistic example		the numer approxima from globa **Limit:** Th finite grid exact answ
8	**Exact inference in structured sequence models** Proposed teaching scope	Derive scaled HMM forward/backward recursions, evidence and filtered/smoothed marginals; establish dense computational cost and distinguish joint path MAP from marginal modes. Give a compact Kalman-filter application of L01 Gaussian conditioning, without a second full sequence implementation.	• 9.2.1–9.2.5 Hidden-state calculations and scaling • 9.2.6 Whole history versus separate state decisions • Selected 8.1.2 / 8.2.2 Gaussian filtering: short application		Implemen marginals and an asy errors. Sin supplied. S to obtain e controllin The main previously not includ
9	**Variational inference: posterior geometry and pathwise gradients** Proposed teaching scope	Reuse L06 ELBO identity; optimize restricted reverse KL and work the correlated-Gaussian diagonal optimum. Derive the total pathwise gradient including direct parameter terms and regularity conditions; apply bounded coordinate-ascent VI and recognize amortization.	• 10.1.1–10.1.2 Fitting a restricted distribution • 6.3.3 / 6.3.5; 10.2.1.1 Derivatives through Gaussian draws • 6.3.4 Another gradient estimator: statement only • 10.3.1 Coordinate updates; 10.1.5 learned inference: introduction only		Fit a diago regression Compare solution; u checks an Separate c and differe correct ful **Limit:** Us economic- replaced; r adaptation
10	**Monte Carlo and MCMC: invariance, mixing and estimator error** Proposed teaching scope	Derive IID estimator variance, stationarity/detailed balance and the general Metropolis–Hastings ratio, retaining rejected states. Explain Gibbs cancellation briefly and dependent-sample error under its conditions. Distinguish invariant transitions, finite-run mixing and function-specific effective sample size; recognize HMC mechanics only.	• 11.2–11.2.2 Averages and sampling error • 2.6.4; 12.2.1–12.2.2 Invariant distribution and acceptance rule • 12.3.1–12.3.2 Conditional-coordinate sampling: brief derivation • Selected 12.6.3 Dependent-sample error; 12.5.3 Hamiltonian mechanics only		Implemen target. Co assess the references query-dep separately for a speci **Limit:** He explained derivative implemen
11	**Neural conditional models: heteroskedastic likelihood and regularization** Proposed teaching scope	Nonlinear feedforward mean/positive-scale model; derive heteroskedastic Gaussian NLL and output derivatives, reusing AD/SGD. Distinguish conditional observation noise from posterior weight uncertainty via total variance. Work the Gaussian-prior/L2 coefficient under mean-loss normalization.	• Selected 16.2.1–16.2.2 / 16.3.1 Small feedforward networks • 15.2.2.1 Gaussian prior and weight shrinkage • 14.2.1 Evaluating predicted probabilities	• 13.2.4.4 Noise that depends on input • 13.5.2 Weight shrinkage; 13.5.1 stopping training: brief mention	Fit one sha held-out N supplied fi law. Matc training da conditiona variance d predictive describe o They do n One CPU course ada
12	**Variational autoencoders: amortized inference and rate–distortion** Proposed teaching scope	Generative model versus amortized posterior; joint encoder/decoder objective with event-summed Gaussian NLL and diagonal KL, then observation averaging. Reuse ELBO/pathwise/neural results. Derive rate as mutual information plus aggregate-posterior mismatch; interpret beta, collapse and nonidentifiability.	• Selected 21.1–21.2 Model, inference network and training objective • 21.3.1 Changing the penalty weight • Selected 5.4.2 Reconstruction/information tradeoff • Selected 21.4 When the model ignores hidden coordinates		Complete joint fit. U comparis average pr separate. S with a neu rate tradeo or latent p source. **Li** coordinate runtime re identified unrun.
13	**Project presentations and course synthesis**	Student project presentations connecting probabilistic assumptions, computation, prediction, evaluation, uncertainty and scientific interpretation. No additional substantive lecture topic is committed at this stage.			No notebo

Deliberate scope boundaries

This course develops selected mathematical capabilities, not an exhaustive survey of Murphy. The exclusions below are deliberate; they are not hidden promises to cover material during L13 project presentations. The independent architecture audit evaluates source coverage and teaching feasibility.

Other possible topic	Choice in this plan	What we give up
More advanced optimization	Basic derivatives and small-batch updates are sufficient here	No separate study of curved-geometry updates, expensive black-box search, transport or subset selection.
Particle methods for changing hidden states	Exact sequence calculations in L08 and sampling theory in L10	No method for tracking a complex changing state with a cloud of weighted samples (Book 2 Ch. 13).
MRI classification and continuous forecasting	Missing-data fitting and discrete hidden-state lab selected	Less biomedical and forecasting practice. Either application can replace its corresponding lab after review.
More complex regression and prediction checks	Common models in L05 and one neural probability model in L11	No full sparse/hierarchical regression, calibration or guaranteed prediction-interval course.
Larger inference toolkits	One checked method per main lab	No general-purpose graph solver, broad variational-family catalogue or full Hamiltonian sampler derivation.
Bayesian neural networks and Gaussian processes	Small fitted networks only	No separate treatment of uncertainty about network weights or Gaussian-process prediction.
Other generative models	A variational autoencoder in L12	No flow, diffusion or adversarial model teaching. These are different ways to learn data distributions and need their own time.